# CH 3 NPUT-OUTPUT RELATIONSHIP

## 3.1 THE LAW OF DIMINISHING RETURNS

### MARKING SCHEME

| | | | | |
|---|---|---|---|---|
| 1990/CE/II/06<br>C | 1996/CE/II/22<br>D | 2002/CE/II/13<br>C (58%) | 2006/CE/II/17<br>A (58%) | 2014/DSE/I/05<br>C (79%) |
| 1992/CE/II/17<br>C | 1996/CE/II/23<br>B | 2003/CE/II/20<br>C (62%) | 2008/CE/II/16<br>D (53%) | 2015/DSE/I/09<br>A (79%) |
| 1994/CE/II/21<br>C | 1999/CE/II/15<br>D | 2004/CE/II/18<br>C (69%) | 2010/CE/II/19<br>C (44%) | 2018/DSE/I/10<br>C (65%) |

**2025/DSE/I/06**

B

*Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.*

1994/CE/I/11(a)
Law of diminishing returns:
When variable factors are added continuously to the fixed factors, the marginal product will eventually diminish. (3)

| Fertilizers (Units) | Land (Units) | Labour (Units) | TP (Units) | MP (Units) |
|---|---|---|---|---|
| 0 | 10 | 40 | 30 | - |
| 1 | 10 | 40 | 70 | 40 |
| 2 | 10 | 40 | 120 | 50 |
| 3 | 10 | 40 | 160 | 40 |
| 4 | 10 | 40 | 190 | 30 |

From the above table, after 2 units of the variable factor (fertilizers) are employed, an additional unit of fertilizers leads to a fall in MP ⇒ the law applies to the farm. (3)

1997/CE/I/3
Assume that the factory has 100 workers. (The number is arbitrary.) After 95 workers are employed, the change in MP caused by adding more and more workers in production can be found by converting the table given in the question into the following table:

| No. of workers coming to work | MP (units / day) |
|---|---|
| 95 | - |
| 96 | 60 |
| 97 | 50 |
| 98 | 40 |
| 99 | 30 |
| 100 | 20 |

(2)

In other words, when the variable factor (labour) is continuously added to the fixed factors (land, capital and entrepreneurship), the marginal product will eventually diminish. This is the law of diminishing marginal returns. (3)

1998/CE/I/4
(a) When a variable factor is successively added to the fixed factor(s), (holding technology constant) (2)
the marginal product will eventually diminish. (2)

(b) No, because
all factors are variable / long-run production. (2)

2000/CE/I/4

The law states that when variable factors are added continuously to the fixed factors, (holding technology constant), the marginal product will eventually decrease. (3)

The marginal product calculated is shown below: (3)

| AP | MP |
|---|---|
| 30 | - |
| 34 | 62 |
| 37 | 61 |
| 39 | 57 |
| 40 | 50 |

∴ The data illustrate the law of D.M.P.

2001/CE/I/3

(The data show that) machinery is a fixed factor and (1)
when the variable factor, labour, is added successively to machinery, (1)
the marginal product eventually diminishes (2)

| Labour (units) | Machinery (units) | MP (units) |
|---|---|---|
| 10 | 5 | - |
| 11 | 5 | 50 |
| 12 | 5 | 100 |
| 13 | 5 | 150 |
| 14 | 5 | 130 |

(2)

∴ the data illustrate the law of diminishing marginal returns. (1)

2003/CE/I/2

(a) The law states that as the variable factor is continuously added to the fixed factor, the marginal product will eventually diminish. (3)

(b)

| Machine (Unit) | MP (Unit) |
|---|---|
| 1 | - |
| 2 | 14 |
| 3 | 16 |
| 4 | 10 |

It illustrates the law in (a) since MP eventually diminishes from 16 units of output to 10 units of output when the 4th unit of machine is added to the production. (3)

2006/CE/I/4

The law states that when variable factors are added continuously to the fixed factors, [holding technology constant,] the marginal product will eventually decrease. (3)

| Labour (Units) | 4 | 5 | 6 |
|---|---|---|---|
| Marginal Product (Units) | 60 | 30 | 20 |

(2)

**OR**

Diminishing marginal product of labour sets in when labour increases from 4 units to 5 units as the marginal product falls from 60 to 30 units. (2)

∴ The data illustrate the law of diminishing marginal returns. (1)

2007/CE/I/3

(a) The law states that when variable factors are added continuously to the fixed factors, the marginal product will eventually decrease. (3)

(b)

| Labour (units) | Marginal product (units) |
|---|---|
| 5 | - |
| 6 | 220 |
| 7 | 210 |
| 8 | 170 |

(2)

Yes, because (1)
the marginal product decreases. (1)

2009/CE/I/3

The law states that when variable factors are added continuously to the fixed factors, the marginal product will eventually decrease. (3)

| Machine (units) | Marginal output of labour (units) |
|---|---|
| 3 | 18 |
| 4 | 22 |
| 5 | 21 |

(2)

Yes, because (1)
the marginal product decreases .

2012/DSE/II/2

The law states that (holding technology constant,) when more units of a variable factor are added successively to a given quantity of fixed factors, the marginal product of this factor will eventually diminish. (3)

No, because (1)
there is no fixed factor / long-run production. (1)

2016/DSE/II/3

The law states that (holding technology constant) when more units of a variable factor are added successively to a given quantity of fixed factors, the marginal products of this factor will eventually diminish. (3)

Yes, because (1)
when labour increases from 3 to 4 units, the marginal product of labour drops from 18 to 14 units (, so the law of diminishing returns holds). (2)

2021/DSE/II/1c

(c) **Law of diminishing returns. When a larger amount of variable factor are continuously added to a given amount of a fixed factor, the marginal product (of the variable factor) would eventually diminish, holding technology constant. In this case, the rider is the variable factor and the firm's operating system is the fixed factor.** [2] **(5)**

**2023/DSE/II/3**

| | Marks |
|---|---|
| (a) Its quantity does not vary with output. | (1) |
| (b) Yes, because | (1) |
| the marginal product of labour drops from 48 to 41 units when the quantity of labour (a variable factor) increases from 12 to 13 units, given a fixed quantity (12 units) of machinery (fixed factor). | (2) |

**2025/DSE/II/1**

| | |
|---|---|
| The law states that (holding technology constant), when more units of a variable factor are added successively to a given quantity of fixed factors, the marginal product of this factor will eventually decrease. | (2) |
| Yes, because given a constant number of machines, a fixed factor, when labour increases from 3 units to 4 units, the marginal product of labour drops from 25 units to 23 units. | (3) |